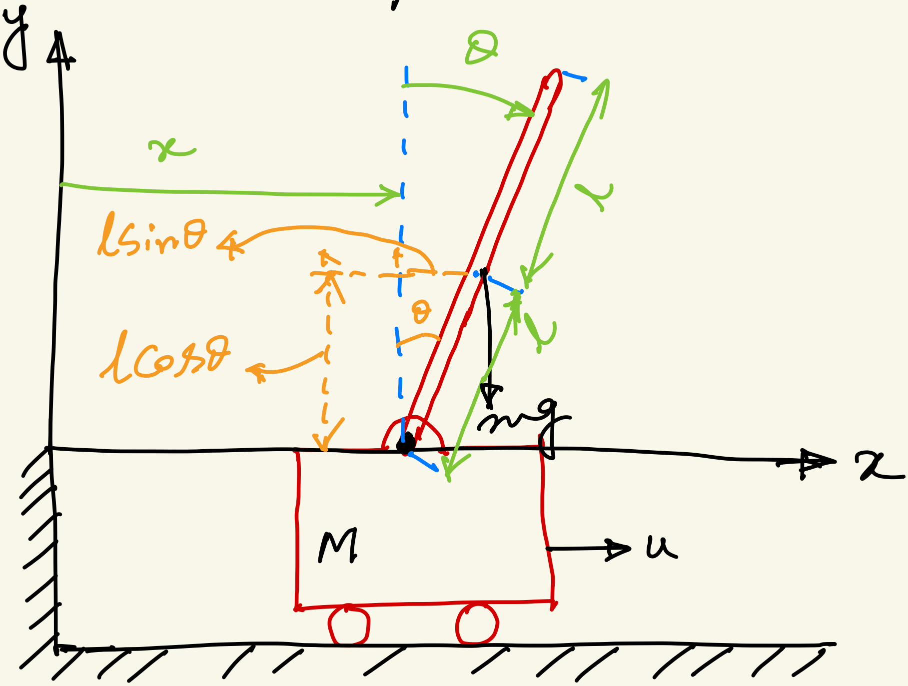
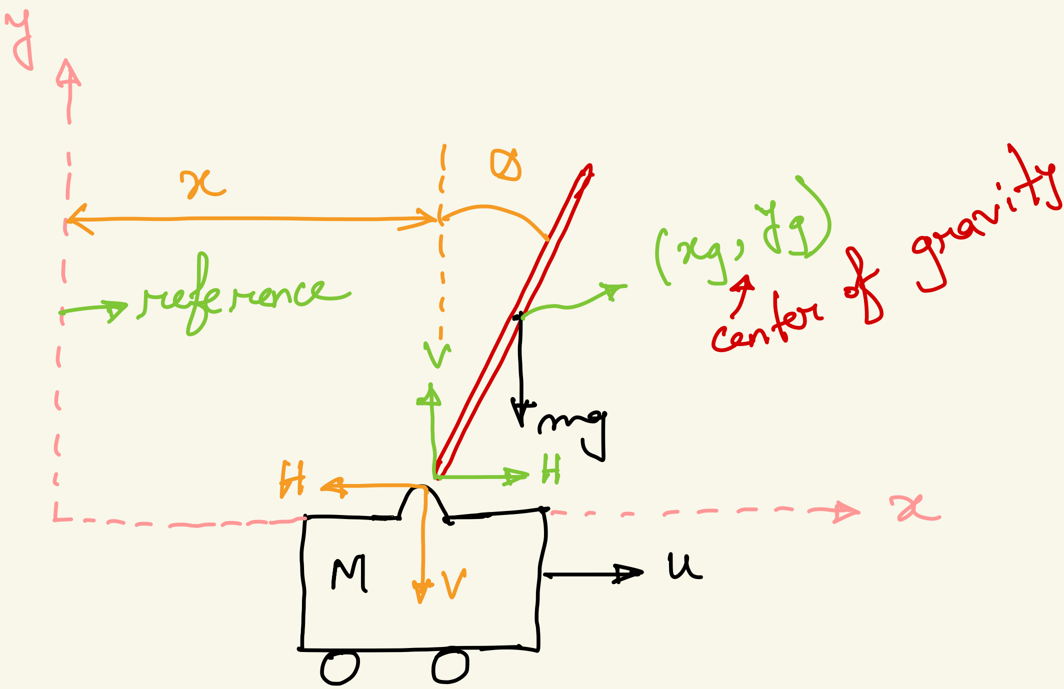
Inverted pendulum on a cart





The center of gravity of the pendulum rod can be defined with respect to the reference as follows:

$$x_g = x + l\sin\theta$$

$$y_g = l\cos\theta$$

The pendulum rod has rotational motion as well as translational motion. Taking moments about the center of gravity:

$$I\ddot{\theta} = Vl\sin\theta - Hl\cos\theta$$

I = moment of inertia of the rod

about the center of gravity.

Balancing forces in the x-direction

$$m\frac{d^2}{dt^2}(x + l\sin\theta) = H$$

Balancing forces in the y-direction:

$$m\frac{d^2}{dt^2}(l\cos\theta) = V - mg$$

The cart moves only in the x-direction. Balancing forces:

$$M\frac{d^2x}{dt^2} = u - H$$

Since we would like to keep the

pendulum vertical, so $\theta(t)$ is small. And $\dot{\theta}(t)$ is also small. We can make the following assumptions:

$$\sin\theta = \theta$$

$$\cos\theta = 1$$

$$\theta\dot{\theta}^2 = 0$$

we get:

$$I\ddot{\theta} = Vl\theta - Hl$$

$$m(\ddot{x} + l\ddot{\theta}) = H$$

$$0 = V - mg \Rightarrow V = mg$$

We substitute the value of H

$$M\frac{d^2x}{dt^2} = u - H$$

$$= u - m(\ddot{x} + l\ddot{\theta})$$

$$= u - m\ddot{x} - ml\ddot{\theta}$$

$$\Rightarrow M\ddot{x} = u - m\ddot{x} - ml\ddot{\theta}$$

$$\Rightarrow (M+m)\ddot{x} + ml\ddot{\theta} = u \quad \text{——} (i)$$

$$I\ddot{\theta} = mgl\theta - l(m\ddot{x} + ml\ddot{\theta})$$

$$= mgl\theta - ml\ddot{x} - ml^2\ddot{\theta}$$

$$\Rightarrow (I + ml^2)\ddot{\theta} + ml\ddot{x} = mgl\theta \quad \text{——} (ii)$$

Equation (i) and (ii) constitute the

equation of motion for a pendulum on a cart. In order to obtain the state-space form, we need to further simplify the equation (I) and (II).

The moment of inertia of the rod is small. $I = 0$.

$$\left(\cancelto{0}{I} + ml^2\right)\ddot{\theta} + ml\ddot{x} = mgl\theta$$

$$\Rightarrow ml^2\ddot{\theta} + ml\ddot{x} = mgl\theta$$

$$\Rightarrow ml\ddot{x} = mgl\theta - ml^2\ddot{\theta}$$

$$\Rightarrow \ddot{x} = g\theta - l\ddot{\theta}$$

and $$\ddot{\theta} = \frac{g}{l}\theta - \frac{1}{l}\ddot{x}$$

$$(M+m)(g\theta - l\ddot{\theta}) + ml\ddot{\theta} = u$$

$$\Rightarrow (M+m)g\theta - (M+m)l\ddot{\theta} + ml\ddot{\theta} = u$$

$$\Rightarrow Mg\theta + mg\theta - Ml\ddot{\theta} - \cancel{ml}\ddot{\theta} + \cancel{ml}\ddot{\theta} = u$$

$$\Rightarrow Ml\ddot{\theta} - (M+m)g\theta = -u$$

$$\Rightarrow Ml\ddot{\theta} = (M+m)g\theta - u \quad \text{--- } (3)$$

$$(M+m)\ddot{x} + ml\left(\frac{g}{l}\theta - \frac{1}{l}\ddot{x}\right) = u$$

$$\Rightarrow (M+m)\ddot{x} + mg\theta - m\ddot{x} = u$$

$$\Rightarrow M\ddot{x} + \cancel{m\ddot{x}} + mg\theta - \cancel{m\ddot{x}} = u$$

$$\Rightarrow M\ddot{x} = u - mg\theta \qquad (4)$$

Equation (3) and (4) is our desired form.

$$x_1 = \theta \qquad \dot{x}_1 = \dot{\theta} = x_2$$

$$x_2 = \dot{\theta} \qquad \dot{x}_2 = \ddot{\theta} = \frac{M+m}{Ml} g\theta - \frac{u}{Ml}$$

$$x_3 = x \qquad \dot{x}_3 = x_4$$

$$x_4 = \dot{x} \qquad \dot{x}_4 = \ddot{x} = -\frac{mg}{M}\theta + \frac{u}{M}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \left(\frac{M+m}{Ml}\right)g & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{mg}{M} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{Ml} \\ 0 \\ \frac{1}{M} \end{bmatrix} u$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

This is the desired state-space form of the pendulum on a cart.